

37-LB: Diabetic Retinopathy (DR) Screening Performance of an Artificial Intelligence (AI) System: Analysis from a Pivotal Multicenter Prospective Clinical Trial

BRUCE W. BODE; ELI IPP; VIRAL N. SHAH; SRINIVAS R. SADDA



Diabetes 2019;68(Supplement_1):37-LB

<https://doi-org.proxy1.library.jhu.edu/10.2337/db19-37-LB>



Split-Screen



Views 



Share 



Cite



Get Permissions

A prospective study (NCT03112005) evaluated DR screening using the EyeArt AI eye screening system against the ETDRS clinical reference standard (CRS). A total of 893 subjects with diabetes were enrolled at 15 U.S. clinical centers and completed 2-field fundus photography for analysis by the EyeArt system and dilated 4-wide field stereo fundus photography for adjudicated expert grading to establish CRS.

The EyeArt system provided eye-level results for referable DR (rDR) [moderate non-proliferative DR (NPDR) or higher on the International Clinical DR (ICDR) severity scale or clinically significant diabetic macular edema (CSDME)] and for vision-threatening DR (vtDR) [severe NPDR or higher on the ICDR scale or CSDME]. Out of 1718 eyes with gradable CRS for rDR, 1392 eyes were negative and 326 were positive. Out of 1702 eyes with gradable CRS for vtDR, 1588 were negative and 114 were positive.

The EyeArt system was evaluated against CRS for identifying rDR and vtDR under an undilated protocol and a dilate-if-needed protocol (where dilated images were used for subjects with ungradable EyeArt results on undilated images). Results (Table) indicate that the EyeArt AI system compares favorably with the clinical reference standard and meets the predetermined study endpoints for the detection of referable and vision-threatening DR in people with diabetes.

[View large](#)

[Download slide](#)

Disclosure

B.W. Bode: Consultant; Self; ADOCIA, Lexicon Pharmaceuticals, Inc., Novo Nordisk Inc. Research Support; Self; Becton, Dickinson and Company, Dexcom, Inc., Diasome Pharmaceuticals, Inc., Eli Lilly and Company, Eyenuk Inc., Insulet Corporation, National Institutes of Health, Novo Nordisk Inc., Sanofi Research & Development, Senseonics, Xeris Pharmaceuticals, Inc. Speaker's Bureau; Self; AstraZeneca, Boehringer Ingelheim Pharmaceuticals, Inc., Janssen Pharmaceuticals, Inc., Novo Nordisk Inc., Sanofi US, Senseonics. Stock/Shareholder; Self; AgaMatrix, Glytec, LLC. **E. Ipp:** Research Support; Self; Eyenuk Inc., Novartis Pharmaceuticals Corporation. **V.N. Shah:** Advisory Panel; Self; Sanofi US. Consultant; Self; Dexcom, Inc. **S.R. Sadda:** Consultant; Self; Carl Zeiss Meditec, Centervue, Heidelberg Engineering, Optos. Other Relationship; Self; Nidek. Other Relationship; Spouse/Partner; Topcon.

Funding

[Skip to Main Content](#)

National Institutes of Health (FY027241, FY026864)

This site uses cookies. By continuing to use our website, you are agreeing to [our privacy policy](#).

[Accept](#)

Readers may use this article as long as the work is properly cited, the use is educational and not for profit, and the work is not altered. More information is available at

<http://www.diabetesjournals.org.proxy1.library.jhu.edu/content/like>



[View Metrics](#)

Email Alerts

[Article Activity Alert](#)

[Online Ahead of Print Alert](#)

[Latest Issue Alert](#)

Most Read

Most Cited

[300-OR: Beneficial Effect of the Combination Therapy of Cagrilintide, a Dual Amylin/Calcitonin Agonist, and Tirzepatide, a Dual GLP-1/GIP Agonist, on Body Weight Loss in Obese Rats](#)

[Subtypes of Type 2 Diabetes Determined From Clinical Parameters](#)

[Bioengineered Artificial Extracellular Vesicles Presenting PD-L1 and Gal-9 Ameliorate New-Onset Type 1 Diabetes](#)

[The Role of GIP in the Regulation of GLP-1 Satiety and Nausea](#)

[Cushing Syndrome, Hypercortisolism, and Glucose Homeostasis: A Review](#)

Online ISSN 1939-327X Print ISSN 0012-1797

[Skip to Main Content](#)

This site uses cookies. By continuing to use our website, you are agreeing to [our privacy policy](#).

Accept

Journals

Diabetes
Diabetes Care
Diabetes, Obesity, and
CardioMetabolic CARE
Clinical Diabetes
Diabetes Spectrum
Standards of Medical
Care in Diabetes
Scientific Sessions
Abstracts
BMJ Open Diabetes
Research & Care

Podcasts

Diabetes Core Update
Diabetes Care "On Air"
DiabetesBio
Diabetes Day by Day
Diabetes Core Update:
Special Editions
The Points of CARE

Other

Special Collections
DiabetesPro®
Diabetes Food Hub™
Know Diabetes By
Heart™

Resources

ADA Professional Membership
ADA Member Directory
Diabetes.org

Books

ShopDiabetes.org
ADA Professional Books

Clinical Compendia

Clinical Compendia
Home

News

Latest News
DiabetesPro SmartBrief

About

About the ADA
Journal Policies
Accessibility
For Reviewers
Advertising in ADA
Journals
Reprints and Permission
for Reuse
Copyright Notice/Public
Access Policy
Privacy Policy

